

StudyFlix Development Report

Date: 03 September 2026

Developer: U Mulaudzi

Project: StudyFlix Android Application

Overview

Today's session focused primarily on the **Assignments feature**, including Firebase integration, submission functionality, navigation restructuring, UI redesign, debugging, and aligning the feature with the StudyFlix architecture.

A complete assignment workflow was successfully implemented:

Assignments

↓

Open Assignment

↓

View Details

↓

Read Instructions

↓

Continue To Assignment

↓

Answer Questions

↓

Submit Assignment

↓

Save To Firebase

1. Assignment Feature Development

Assignment Details Screen

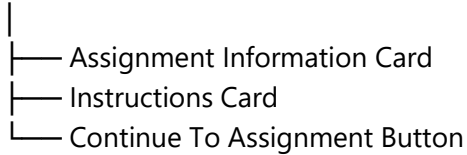
Implemented a dedicated Assignment Details screen displaying:

- Assignment Title
- Subject
- Teacher
- Examiner

- Total Marks
- Duration
- Due Date
- Status
- Instructions

UI Structure

Assignment Details



2. Assignment Question Screen

Created a dedicated Assignment Question Screen.

Architecture Change

Previous Design

Assignment Details
Instructions
Questions
Answers
Submit

New Design

Assignment Details
↓
Continue To Assignment
↓
Assignment Questions Screen
↓
Submit Assignment

This separation greatly improved usability and reduced screen clutter.

3. Assignment Navigation

Implemented a new assignments navigation flow.

Added

AssignmentDetails Route
AssignmentQuestions Route

Working Flow

Assignment List
↓
Assignment Details
↓
Assignment Questions

Navigation is functioning correctly with assignment IDs being passed between screens.

4. Firebase Assignment Loading

Connected Assignment screens to Firestore.

Completed

- AssignmentRepository
- AssignmentRepositoryImpl
- AssignmentDetailsViewModel
- Assignment loading by ID
- Question loading

Result

Assignments and questions now load dynamically from Firebase.

5. Question Mapping

Successfully mapped question data from Firestore.

Loaded Data

Question Number

Question Text

Result

Questions display correctly on Android.

6. Answer Capture System

Implemented answer state management.

Added

`mutableStateMapOf<String, String>()`

Result

Students can type answers and inputs persist while navigating through questions.

Example:

Question 1 → 56

Question 2 → 34

Question 3 → 23

7. Assignment Submission System

Created Assignment Submission Version 1.

Created Model

AssignmentSubmission

Containing:

assignmentId

studentId

submittedAt

answers

Repository Layer

Added:

submitAssignment()

to:

AssignmentRepository
AssignmentRepositoryImpl

ViewModel Layer

Implemented:

submitAssignment()

inside:

AssignmentDetailsViewModel

8. Firebase Submission Pipeline

Successfully implemented:

Screen

↓

ViewModel

↓

Repository

↓

Firestore

Debug Log Result

ASSIGNMENT_SUBMIT

Submitting:

AssignmentSubmission(
assignmentId=...
studentId=...

```
answers=...  
)
```

Submission saved successfully

Status

☒ Firebase submission working

9. Firestore Security Debugging

Resolved:

PERMISSION_DENIED

Root Cause

Temporary user ID:

TEMP_USER

did not satisfy Firestore security rules.

Fix

Replaced temporary value with:

```
FirebaseAuth.getInstance()  
.currentUser  
?.uid
```

Result

Firestore accepts submission requests.

10. Firestore Serialization Fix

Resolved Firestore crash:

Maps with non-string keys are not supported

Previous

Map<Int, String>

Updated

Map<String, String>

Result

Submission data is accepted by Firestore.

11. Assignment UI Redesign

Started aligning assignments with the StudyFlix design system.

Added

Assignment Details Screen

Top App Bar
Back Button
Background Color
Rounded Cards
Card Spacing

Assignment Question Screen

Top App Bar
Back Button
Assessment Summary Card
Submit Section

12. Assessment Summary

Added summary information to Question Screen.

Displays:

Total Questions

Total Marks

Duration

Example

Assessment Summary

Total Questions: 6

Total Marks: 50

Duration: 60 Minutes

13. Submission State Handling

Added submission feedback states.

Added

submitted

submitting

UI Feedback

Submitting Assignment...

and

☒ Assignment Submitted

14. Question Screen Layout Refactoring

Began restructuring the Question Screen.

Goal

Header

Assessment Summary

Scrollable Questions Section

Submit Card

instead of scrolling the entire screen.

Progress

- Introduced LazyColumn
- Began fixed-header architecture
- Began fixed submit panel architecture

Further refinement still required.

15. Compose Dependency Investigation

Discovered build issues caused by:

Compose Foundation 1.12.0

which required:

AGP 9.1+

Compile SDK 37+

while project currently uses:

AGP 8.6.1

Compile SDK 35

Root Cause

Located in:

libs.versions.toml

foundation = "1.12.0"

Issue identified for correction.

Current Assignment Status

Functional

- ☒ Assignment List
- ☒ Assignment Details
- ☒ Instructions

- ✓ Question Loading
 - ✓ Answer Entry
 - ✓ Navigation
 - ✓ Firebase Submission
 - ✓ Back Navigation
 - ✓ Assessment Summary
 - ✓ Submission Status Messages
-

UI In Progress

- ⚠ Text visibility improvements
 - ⚠ Question card styling
 - ⚠ Question-only scrolling
 - ⚠ Typography improvements
 - ⚠ Color consistency
-

Next Session Plan

UI Completion

Assignment Details

Center Assignment Title
Improve Information Text Visibility
Improve Card Typography
Finalize Styling

Question Screen

Fix Question Text Visibility
Finalize Scrollable Question Area
Improve Question Card Design
Improve Submit Section

Functionality Improvements

Prevent Duplicate Submissions
Check Existing Submission
Show Completed Status
Disable Resubmission

Future Enhancements

Assignment Results
Marks Display
Memo Viewing
Submission History
Grading Integration

GitHub Commit

```
git add .  
git commit -m "Implemented assignment workflow, Firebase submissions, navigation refactor, UI redesign and Firestore fixes"  
git push origin main
```

Major Achievement of the Day

Successfully completed the first end-to-end Assignment workflow:

```
Student  
↓  
Opens Assignment  
↓  
Reads Instructions  
↓  
Answers Questions  
↓  
Submits Assignment  
↓  
Data Saved
```

☑ Assignment system is now functionally operational from start to finish. 🚀💜